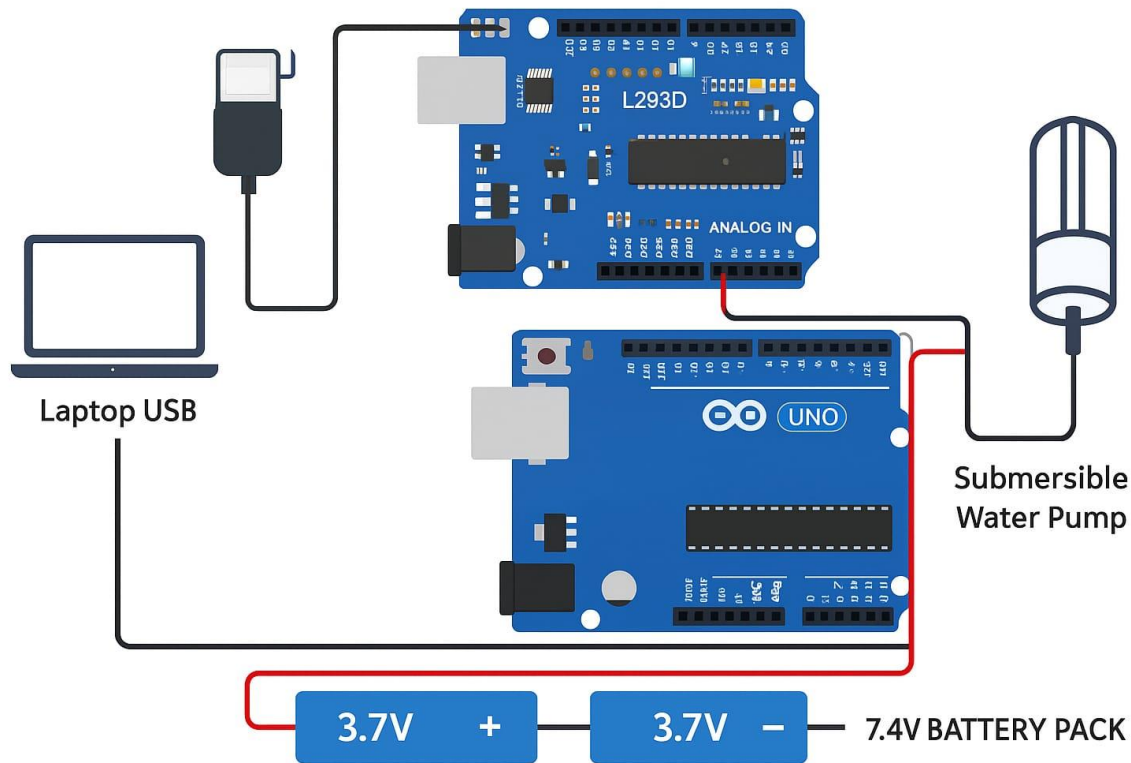


IMPLEMENTATION AND RESULT

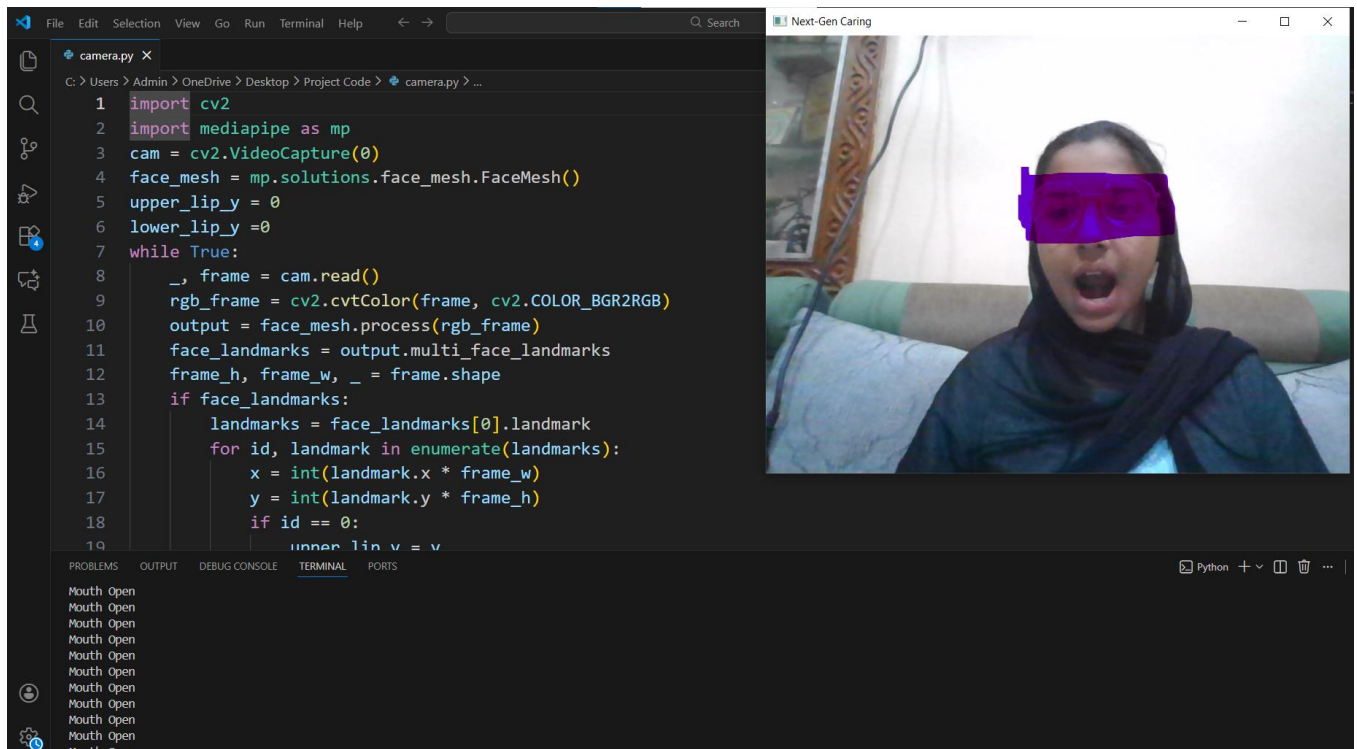
Circuit Setup:



This implemented system uses an Arduino Uno Rev3 to control a submersible water pump with the help of an L293D Motor driver shield. The Arduino is connected to a laptop via USB Type B for programming and communication. A 7.4V battery pack, made by connecting two 3.7V batteries in series, provides the required power to operate the water pump. Since the Arduino cannot drive the pump directly, the L293D motor driver is used as an interface to handle higher current and voltage safely. The motor driver receives control signals from the Arduino and switches the pump ON or OFF accordingly. This setup ensures reliable and safe operation of the pump and is used in our project to feed the Disable Patient without using Hands.

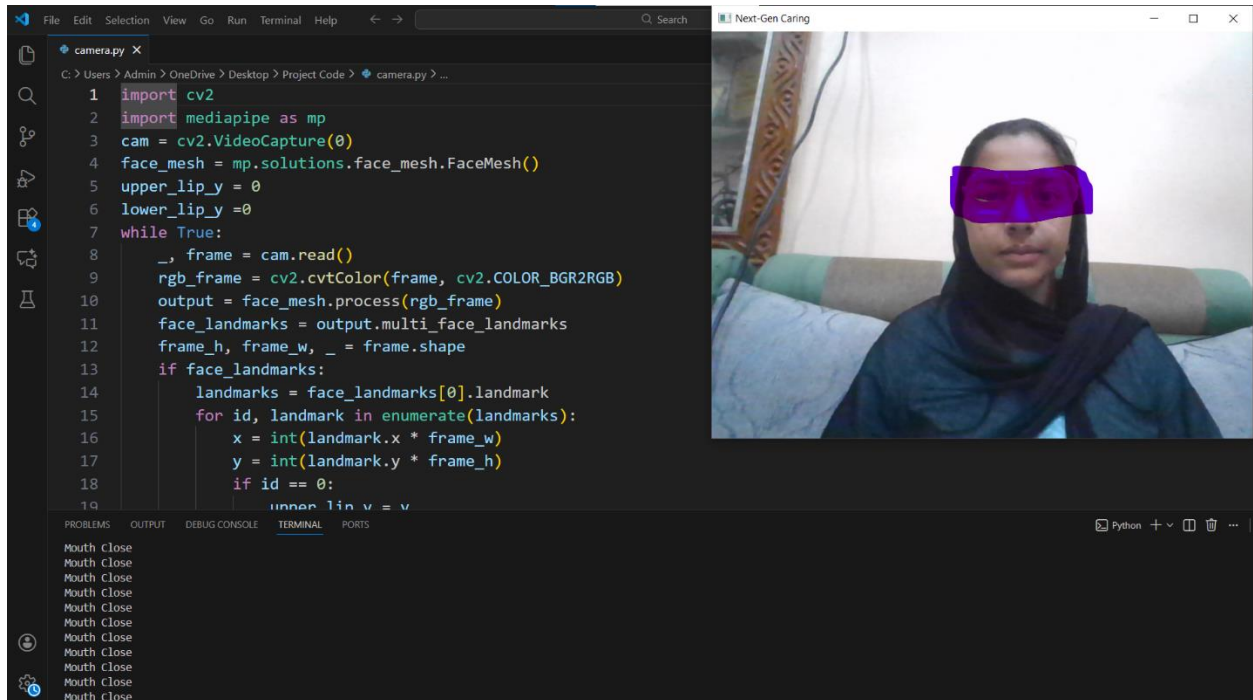
Note: In our project we Mounted the L293D Motor shield on top of the Arduino UNO REV3, for the reliable, weightless circuit..

Mouth Position: OPEN



When the mouth is detected as open, the distance between the upper and lower lip landmarks exceeds the predefined threshold, indicating an intentional feeding request by the patient. The program displays “Mouth Open” and sends a control signal to the microcontroller. After verifying safety conditions, the microcontroller activates the motor-driven pump, allowing liquid to be delivered through the feeding pipe in a controlled manner.

Mouth Position: CLOSE



When the mouth is detected as closed, the system continuously monitors the facial landmarks but does not initiate any action. The distance between the upper and lower lip landmarks remains below the set threshold, so the program displays “Mouth Close” and keeps the motor and pump in an OFF state. This prevents accidental or unsafe liquid dispensing and ensures the system waits for a valid feeding request.

Final Result of Demonstration



During the experimental demonstration, the system continuously monitors the patient's facial movements using a camera connected to the laptop. The real-time video feed is processed using a Python-based computer vision algorithm to detect the position of the patient's mouth. When the mouth is identified as open, the system interprets this as a feeding request. A control signal is then transmitted from the laptop to the Arduino microcontroller through a USB connection. The Arduino, with the motor shield mounted on top, activates the submersible water pump powered by a 7.4V battery.

As a result, water flows through the connected feeding pipe and is delivered to the patient in a controlled manner. The flow continues only while the mouth remains open. Once the mouth closes or the detection signal is withdrawn, the pump is immediately stopped. This real-time interaction demonstrates the successful integration of mouth position detection with automated water feeding, ensuring both responsiveness and patient safety..

